

## DDL – DROP Command

### ❏ What is the function of the DROP command in SQL?

The **DROP** command is a **DDL (Data Definition Language)** command used to permanently remove database objects such as:

- Tables
- Databases
- Views
- Indexes

#### Syntax:

```
DROP TABLE table_name;
```

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### ❏ What are the implications of dropping a table?

- Entire table structure is deleted.
- All data inside the table is permanently removed.
- Indexes, constraints, and triggers on that table are also deleted.
- Cannot be recovered unless backup exists.

⚠ It is a permanent operation.

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## Data Manipulation Language (DML)

DML commands are used to manipulate data inside tables.

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### ❏ Define INSERT, UPDATE, DELETE

#### ✓ INSERT

Used to add new records into a table.

```
INSERT INTO table_name VALUES (...);
```

#### ✓ UPDATE

Used to modify existing records.

```
UPDATE table_name SET column=value WHERE condition;
```

## ✓ DELETE

Used to remove records from a table.

```
DELETE FROM table_name WHERE condition;
```

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## 2 Importance of WHERE clause in UPDATE and DELETE

- Prevents updating or deleting all rows accidentally.
- Specifies which records should be modified.
- Without WHERE → affects entire table.

Example:

```
DELETE FROM students;    -- Deletes ALL records
```

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# Data Query Language (DQL)

## 1 What is SELECT?

The **SELECT** statement is used to retrieve data from a database.

```
SELECT column_name FROM table_name;
```

It can retrieve:

- Specific columns
  - All columns (\*)
  - Filtered data
  - Sorted data
- 

## 2 ORDER BY and WHERE Clause

## ✓ WHERE

Filters records based on condition.

```
SELECT * FROM students WHERE marks > 60;
```

### ✓ **ORDER BY**

Sorts results in ascending (ASC) or descending (DESC).

```
SELECT * FROM students ORDER BY marks DESC;
```

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## **Data Control Language (DCL)**

### **1 Purpose of GRANT and REVOKE**

These commands control user permissions.

#### ✓ **GRANT**

Gives privileges to users.

```
GRANT SELECT ON table_name TO user;
```

#### ✓ **REVOKE**

Removes privileges.

```
REVOKE SELECT ON table_name FROM user;
```

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### **2 Managing Privileges**

Privileges like:

- SELECT
- INSERT
- UPDATE
- DELETE
- ALL

Admins manage access using GRANT and REVOKE.

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## **Transaction Control Language (TCL)**

## 1 Purpose of COMMIT and ROLLBACK

### ✓ COMMIT

Saves changes permanently.

```
COMMIT;
```

### ✓ ROLLBACK

Undo changes before commit.

```
ROLLBACK;
```

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## 2 Transaction Management

A transaction:

- Begins automatically.
  - Ends with COMMIT or ROLLBACK.
  - Ensures data consistency (ACID properties).
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## SQL Joins

### 1 What is JOIN?

JOIN combines data from multiple tables using related columns.

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#### Types of Joins:

| Join Type       | Description                         |
|-----------------|-------------------------------------|
| INNER JOIN      | Returns matching rows               |
| LEFT JOIN       | All rows from left table + matches  |
| RIGHT JOIN      | All rows from right table + matches |
| FULL OUTER JOIN | All rows from both tables           |

Example:

```
SELECT *  
FROM students  
INNER JOIN marks  
ON students.id = marks.id;
```

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## 2 Why Use Joins?

- Combine related data
  - Normalize databases
  - Reduce redundancy
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# SQL GROUP BY

## 1 What is GROUP BY?

Groups rows having same values.

Used with aggregate functions:

- COUNT()
- SUM()
- AVG()
- MAX()
- MIN()

Example:

```
SELECT department, COUNT(*)  
FROM employees  
GROUP BY department;
```

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## 2 GROUP BY vs ORDER BY

| GROUP BY             | ORDER BY                 |
|----------------------|--------------------------|
| Groups rows          | Sorts rows               |
| Used with aggregates | Used for sorting results |

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# SQL Stored Procedure

## 1 What is Stored Procedure?

A stored procedure is a precompiled SQL program stored in database.

Example systems:

- Oracle Database
  - MySQL
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## 2 Advantages

- Faster execution
  - Code reusability
  - Better security
  - Reduced network traffic
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# SQL View

## 1 What is a View?

A virtual table based on a SQL query.

```
CREATE VIEW view_name AS  
SELECT column FROM table;
```

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## 2 Advantages

- Security (hide columns)
  - Simplifies complex queries
  - No data duplication
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# SQL Triggers

## 1 What is Trigger?

A trigger is a procedure that automatically executes when:

- INSERT
- UPDATE
- DELETE happens

Types:

- BEFORE Trigger
- AFTER Trigger

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## 2 Insert vs Update vs Delete Triggers

### **Trigger Executes When**

INSERT New row added

UPDATE Row modified

DELETE Row removed

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# Introduction to PL/SQL

## 1 What is PL/SQL?

PL/SQL (Procedural Language SQL) is an extension of SQL used in:

- Oracle Database

It adds:

- Loops
  - Conditions
  - Variables
  - Exception handling
- 

## 2 Benefits

- Faster execution
- Error handling
- Modular programming
- Better performance

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# PL/SQL Control Structures

## 1 What are Control Structures?

They control execution flow.

### IF-THEN

```
IF condition THEN
    statement;
END IF;
```

### LOOP

```
LOOP
    statements;
END LOOP;
```

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## 2 Why Important?

- Handle complex logic
  - Conditional execution
  - Iterative operations
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# SQL Cursors

## 1 What is Cursor?

A cursor is used to process row-by-row data.

### Types:

- Implicit Cursor (automatic)
  - Explicit Cursor (defined by user)
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## 2 When Use Explicit Cursor?



- When processing multiple rows manually
- When fine control is needed

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# SAVEPOINT, ROLLBACK, COMMIT

## What is SAVEPOINT?

SAVEPOINT creates a checkpoint in a transaction.

```
SAVEPOINT sp1;
```

### ROLLBACK to Savepoint:

```
ROLLBACK TO sp1;
```

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## When Use Savepoints?

- Large transactions
  - Partial rollback required
  - Error handling in complex operations
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## Summary

SQL Categories:

| Category | Commands               |
|----------|------------------------|
| DDL      | CREATE, ALTER, DROP    |
| DML      | INSERT, UPDATE, DELETE |
| DQL      | SELECT                 |

|     |                             |
|-----|-----------------------------|
| DCL | GRANT, REVOKE               |
| TCL | COMMIT, ROLLBACK, SAVEPOINT |